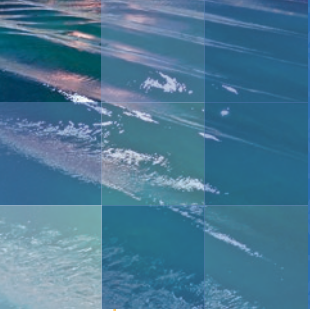




Sinusoidal Family of Functions

The amplitude, period, and vertical shift of a sinusoidal function can be determined from its equation.

Determining Sinusoidal Function Transformations

Sinusoidal functions in the form $f(x) = A \sin Bx + C$ or $f(x) = A \cos Bx + C$ have the following characteristics:

amplitude = $|A|$, where $A \neq 0$

period = $\frac{2\pi}{B}$, where $B > 0$

frequency = $\frac{1}{\text{period}} = \frac{B}{2\pi}$ equation of midline: $y = C$

If $C > 0$, there is a vertical shift up C units.

If $C < 0$, there is a vertical shift down $|C|$ units.

EXAMPLE 1

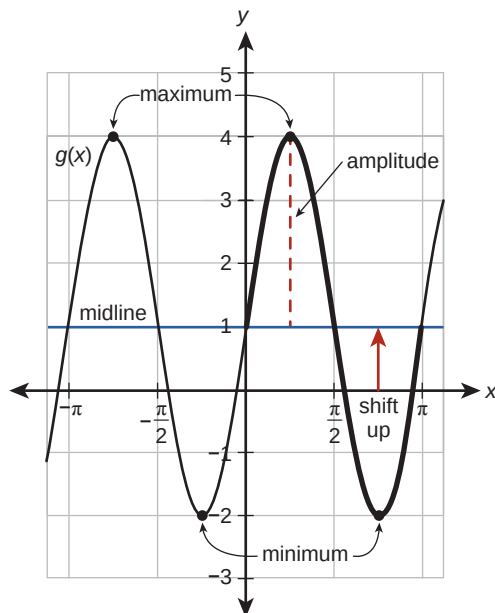
Determine the midline, amplitude, maximum value, minimum value, period, and frequency of the function.

A $g(x) = 3 \sin 2x + 1$

SOLUTION

Since $C = 1$, the equation of the midline is $y = 1$. Since $A = 3$, the amplitude is 3. So the maximum is 3 units above the midline at $y = 4$ and the minimum is 3 units below at $y = -2$.

Since $B = 2$, period $= \frac{2\pi}{2} = \pi$, and frequency $= \frac{1}{\pi}$.

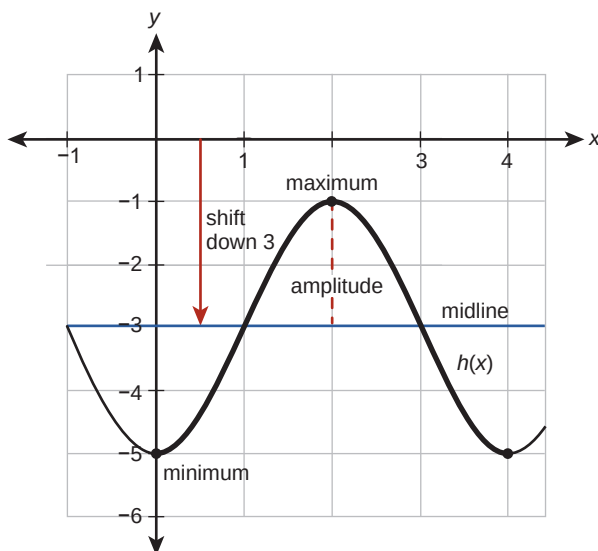


B $h(x) = -2 \cos\left(\frac{\pi x}{2}\right) - 3$

SOLUTION

Since $C = -3$, the equation of the midline is $y = -3$. Since $A = -2$, the amplitude is $|-2| = 2$. Also, since $A < 0$, $h(x)$ is reflected across the midline. The minimum is 2 units below the midline at $y = -5$. The maximum is 2 units above the midline at $y = -1$.

Since $B = \frac{\pi}{2}$, period $= \frac{2\pi}{\frac{\pi}{2}} = 4$, and frequency $= \frac{1}{4}$.



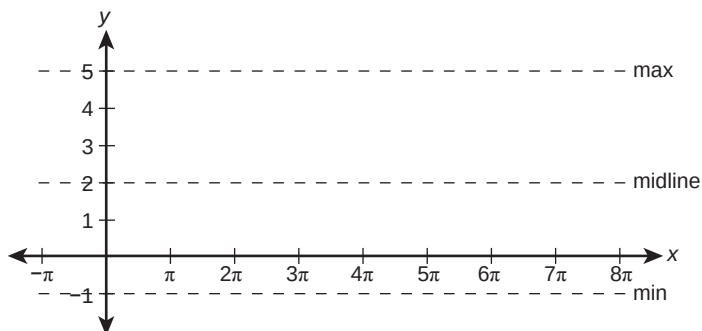
EXAMPLE 2

Graph the function $v(x) = -3 \cos\left(\frac{x}{2}\right) + 2$, and then determine the midline, amplitude, maximum value, minimum value, period, and frequency.

SOLUTION

To graph a sinusoidal function, start by finding the maximum and minimum values.

Since $C = 2$, the equation of the midline is $y = 2$. The amplitude is $|A| = |-3| = 3$. The maximum is 5 and the minimum is -1 . Draw dashed lines for the maximum, minimum, and midline.

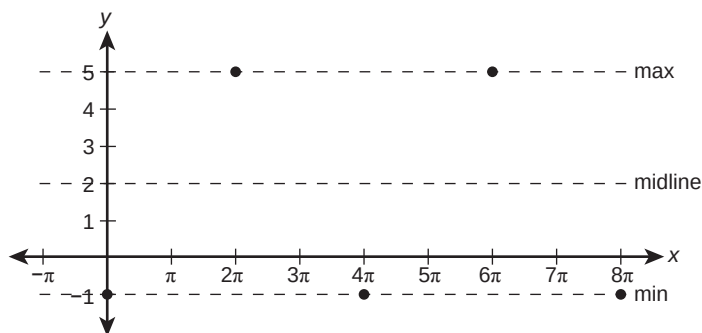


$A < 0$, so the graph is flipped when compared to $y = \cos x$. The period is

$\frac{2\pi}{\frac{1}{2}} = 4\pi$. Start the graph at $x = 0$, and then mark the end of one period at

$x = 4\pi$. The cosine function starts at a maximum or minimum, and this one has $A < 0$, so mark a minimum at $x = 0$, another at $x = 4\pi$, and another at 8π .

Mark maximum values halfway between the minima at $x = 2\pi$ and $x = 6\pi$.



Connect the points with a smooth sinusoidal curve.

